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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	UMvengers
Selected Domain	Domain 2
Selected SDG Track (1–17)	8 and 9
Project / Solution Name	UMvengers-Orbyte
Name (Leader)	Wong Hanz
Email & Phone (Leader)	wonghanz2007@gmail.com 01159470337
University Affiliation(s)	University Malaya
Number of Team Members	Leow Yi Han, Woe Xin Yi, Yeap Yi Theng

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian SMEs contribute almost **40% of Malaysia's GDP and employ nearly half of the national workforce**, making their financial resilience critical to the country's economy. However, delayed customer payments continue to create severe cash-flow pressure. SME Corp Malaysia reported that **39% of MSMEs experience late-payment issues**, while recent industry reports show SMEs continue to face longer payment delays than large corporations, affecting their ability to pay salaries, suppliers, rent and operating expenses.

Beyond debt collection, **overdue invoice management** has become an information and decision-making challenge. Customer conversations, accounting records, company policies and payment histories are scattered across emails, spreadsheets, messaging platforms and ERP systems. Staff **spend excessive time** searching for the right information, often resulting in inconsistent follow-ups and delayed decisions.

Although AI can automate these tasks, businesses **cannot trust responses** that are generated **without accurate company knowledge**. Likewise, AI agents operating **without embedded business ethics** and Malaysian regulatory guidelines may produce actions that expose organisations to compliance and reputational risks. Running multiple AI **agents without intelligent task orchestration** also increases inference costs, making enterprise-grade AI financially inaccessible for many SMEs.

These challenges highlight the need for a solution that can retrieve trusted organisational knowledge, enforce business and government compliance, coordinate specialised AI agents efficiently, and present complex operational workflows through an intuitive visual workspace so finance teams can monitor, understand and manage payment recovery with confidence.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

UMvengers–Orbyte is an AI-powered multi-agent platform that helps Malaysian SMEs manage overdue invoices, understand customer replies, and improve cash-flow visibility through one connected workflow.

Laya acts as the central decision layer, classifying requests and assigning tasks to specialised sub-agents. DeepSeek handles language-intensive tasks, such as understanding payment replies and drafting reminders, while validation and payment-monitoring agents use structured data and deterministic rules. Using LangGraph, the workflow connects these agents and applies **Retrieval-Augmented Generation (RAG)** to an internal company knowledge base built from the LLM Wiki and our Unified Data Source (UDS). This helps agents retrieve relevant company policies, product knowledge, and operational procedures instead of relying only on model memory.

We will also encode **Cradle Fund-aligned business principles and applicable Malaysian laws, regulations, and government ethics requirements** into a dedicated compliance skill for sub-agents. This provides policy checks, escalation rules, and audit records; it supports compliance but does not replace legal review or guarantee it. Human approval remains necessary for sensitive or uncertain actions.

Our target is to keep processing costs below **RM0.10 per request** through lightweight classification, targeted RAG retrieval, and selective use of larger language models. This is a design target that must be validated through actual usage and cost testing.

For the user experience, we propose a **3D interactive platform inspired by the spatial clarity of Apple Park and the collaborative campus concept of Microsoft**. As a product-management tool, its connected buildings represent departments and agents, while each workspace opens a practical dashboard showing responsibilities, workflow status, approvals, and outcomes. This makes a complex multi-agent system easier for SME users and judges to understand.

We will leverage SDC Phase 1 skills in GenAI APIs, database architecture, workflow orchestration, API integration, and cloud deployment to build and test the prototype.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

UMvengers–Orbyte uses a modular technology stack to automate SME payment follow-ups. The web application is deployed on Render, while PostgreSQL stores invoice records, payment statuses, workflow states, and activity logs. Docling supports invoice document extraction, and Pydantic validates structured data.

Laya serves as the central decision layer, classifying requests and routing tasks to specialised sub-agents. DeepSeek supports customer-reply understanding and message drafting. LangGraph coordinates the workflow, while Retrieval-Augmented Generation (RAG) connects agents to an internal knowledge base built from the LLM Wiki and the Unified Data Source (UDS), helping them retrieve relevant company knowledge and procedures.

A dedicated compliance Skill will encode Cradle Fund-aligned business principles and applicable legal and ethical requirements, with human review for sensitive decisions. The front end will feature a bright, interactive 3D campus inspired by Apple Park and Microsoft campus design concepts, with each workspace opening its corresponding operational dashboard.

SDC sponsor-provided services: We will declare the exact GenAI API or cloud platform used only after confirming it against our team's actual workshop access and implementation. Unused sponsor services will not be presented as integrated.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking principles by focusing on the real challenges SMEs face when managing overdue invoices, tracking payments, and communicating with customers. Our team applied a user-centred design-thinking approach to address the challenges Malaysian SMEs face in managing overdue invoices, tracking payments, and following up

with customers. We first identified the key pain points: fragmented business information, repetitive manual work, missed follow-ups, and limited cash-flow visibility. We then designed UMvengers–Orbyte as a connected automation platform with specialised AI agents responsible for invoice validation, payment monitoring, decision-making, customer communication, and reconciliation.

To improve usability, we designed a central dashboard with clear workflow statuses, accessible information, and human approval for sensitive or uncertain actions. Our proposed 3D interactive campus also provides a visual way to navigate different agent departments, while each department opens a practical operational dashboard.

For Phase 2, we will test the prototype using realistic sample invoices and customer replies, gather user feedback, and refine the interface and workflow. We will evaluate AI classification accuracy, data consistency, approval controls, system reliability, and the target cost of less than RM0.10 per request. We will also strengthen data protection, access permissions, error recovery, and cloud deployment on Render. These steps will help us validate technical feasibility and usability before expanding to more SMEs.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8, Target 8.3 — Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalisation and growth of micro-, small- and medium-sized enterprises (MSMEs). UMvengers–Orbyte supports this target by helping SMEs reduce administrative workload and improve payment follow-up visibility
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The draft cites Experian Malaysia reporting average SME payment delays of 64 days in early 2025, compared with 69 days in 2022. This indicates a continuing payment-delay challenge; however, the figures should be verified against the original publication before submission. Project baseline: Before testing automation, we will record staff time spent on payment follow-up for a fixed set of representative test invoices, including time spent reviewing replies and updating payment records.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: Percentage reduction in staff time spent on payment follow-up, measured in minutes per invoice against the manual-process baseline. Supporting KPIs: (1) percentage of cases routed correctly; (2) percentage of required follow-ups completed; and (3) average processing cost per request in RM. All metrics will use a predefined test set and documented measurement rules.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final prototype presentation, demonstrate a 30% reduction in manual follow-up time compared with the measured baseline. Additionally, target at least 90% correct routing on a predefined test set and processing cost below RM0.10 per request. These are validation targets, not proven results. Report actual results, test-set size, and any limitations at the Phase 2 presentation.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ ***No***✅

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ ***Yes***✅ ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
PostgreSQL	Invoice and payment data storage	PostgreSQL License
DeepSeek API	Customer reply understanding and message drafting	Commercial API; governed by provider terms
Laya	Payment classification and decision support	Verify the specific implementation's license or service terms
Render	Cloud hosting and deployment	Proprietary hosting service; provider terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ **No**✅

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

WONG HANZ

24/10/2026